

Author	Title	Journal	Volume	Number	Pages	Year
Darabundit, Champ C.; Scavone, Gary	Discrete port-Hamiltonian system model of a single-reed woodwind instrument	Frontiers in Signal Processing	5		1519450	2025
Domenighini, P.; Buratti, C.	The influence of thermal-hygrometric properties of air on the tune of pipe-organs	Applied Acoustics	213		109611	2023
Gerasimov, Roman	Linear-acoustic effects of asymmetrical undercutting of tone-holes of woodwind instruments	The Journal of the Acoustical Society of America	156	4	2644–2655	2024
Haidar, Gaby Abou; Zeid Daou, Roy Abi; Moreau, Xavier	Lab ViewSimulation of a Flute Wind Musical Instrument Taking into Consideration the Fractional Behavior of Visco Thermal Losses in the Resonator				103–109	2022
Ikoga, Shiryu; Onomata, Tatsuki; Tabata, Ryoya; Iwagami, Sho; Kobayashi, Taizo; Takahashi, Kin'ya	Numerical study on role of foot of a flue organ pipe: relative phases in oscillations among pipe, foot and jet	Acta Acustica	8		50	2024
Ishida, Fukiko; Samejima, Toshiya	Physical modeling of wind instruments involving three-dimensional radiated sound fields	Acoustical Science and Technology	44	3	247–258	2023
Jiang, Jing; Liu, Jingyu; Li, Zijin; Zhang, Tingyu; Yang, Hong	Analysis on the mechanism of sound production and effects of musical flue pipe	Cognitive Computation and Systems	4	1	77–84	2022
Ma, Penglei; Tian, Hao; Yao, Yang; Liu, Guijie; Wang, Xinbao; Ji, Bo	Analysis of cavitation jet characteristics and erosion behavior of an organ-pipe nozzle under constant input power	Ocean Engineering	346		123696	2026
Polychronopoulos, Spyros; Marini, Dimitra; Bakogiannis, Konstantinos; Kouroupetroglou, Georgios Th.; Psaroudakes, Stelios; Georgaki, Anastasia	Physical Modeling of the Ancient Greek Wind Musical Instrument Aulos: A Double-Reed Exciter Linked to an Acoustic Resonator	IEEE Access	9		98150–98160	2021
Schwär, Simon; Dittmar, Christian; Balke, Stefan; Müller, Meinard	Differentiable Pulsetable Synthesis for Wind Instrument Modeling				14792–14796	2026
Sinjankhom, Tantep; Chivapreecha, Sorawat; Chitanont, Nachanant; Kato, Tomonori	Deep Neural Networks for Sound Synthesis of Thai Duct Flute, Khlui				63–67	2021

Duda, T. F.; Lin, Y.-T.; Newhall, A. E.; Helfrich, K. R.; Lynch, J. F.; Zhang, W. G.; Lermusiaux, P. F. J.; Wilkin, J.	Multiscale multiphysics data-informed modeling for three-dimensional ocean acoustic simulation and prediction	Journal of the Acoustical Society of America	146	3	1996–2015	2019
Batten, P.; Spalart, P.; Terracol, M.	Use of hybrid RANS–LES for acoustic source predictions		9,78052E+12		128–166	2007
Liang, J.	Harmonizing minds and machines: survey on transformative power of machine learning in music	Frontiers in Neurorobotics	17			2023
Wang, H.; Zhai, Z. J.	Advances in building simulation and computational techniques: A review between 1987 and 2014	Energy and Buildings	128		319–335	2016
Villeneuve, J.; Leonard, J.	Mass-interaction physical models for sound and multi-sensory creation: Starting anew				187–194	2019
Porcino, T.; Clua, E.	Challenges in Data Visualization with Extended Reality Devices		15192 LNCS		305–310	2025
Leonard, J.; Villeneuve, J.	Design and implementation of real-time physically-based virtual musical instruments: A balancing act		2020-June		234–241	2020
Villeneuve, J.; Cadoz, C.; Leonard, J.	Analysis of physical models, analysis by means of physical models, sound synthesis and frequency analysis by the mean of cordis-anima masse-interaction networks	Traitement du Signal	32	4	365–390	2015
Kushida, N.; Lin, Y.-T.	Long-Range Hydroacoustic Propagation Modelling Schemes on Distributed Memory Parallel Computers	Journal of Theoretical and Computational Acoustics	33	1		2025
Titze, I. R.; Lucero, J. C.	Voice Simulation: The Next Generation	Applied Sciences (Switzerland)	12	22		2022
Wu, Y.-Z.; Han, J.-H.; Guo, T.-F.; Xu, H.-W.; Zheng, S.-L.; Chen, D.-Y.	Physical modeling of membrane musical instrument	Xitong Fangzhen Xuebao / Journal of System Simulation	18	7	1745–1749+1772	2006
Collins, M. D.; Turgut, A.; Buckingham, M. J.; Gerstoft, P.; Siderius, M.	Selected Topics of the Past Thirty Years in Ocean Acoustics		1-Apr		1:67–1:88	2025

Siebein, K.; Siebein, G.; Roa, M.; Miller, J.; Vetterick, M.	Integrated Architectural and Acoustical Design of Media Recording Studios				514–520	2022
Pluta, M.	Simulation of Infeasible Instruments in a Sound Synthesizer – Implementation and Control	Vibrations in Physical Systems	33	2		2022
Zhang, L.; Da, L.; Zhou, Y.	A new ocean acoustic model in computational oceanographic physics					2009
Lermusiaux, P. F. J.; Haley, P. J.; Mirabito, C.; Ali, W. H.; Bhabra, M.; Abbot, P.; Chiu, C.-S.; Emerson, C.	Multi-resolution Probabilistic Ocean Physics-Acoustics Modeling: Validation in the New Jersey Continental Shelf					2020
Pfeifle, F.; Bader, R.	Real-time finite difference physical models of musical instruments on a Field Programmable Gate Array (FPGA)					2012
Kailas, G.; Tiwari, N.	Design for Immersive Experience: Role of Spatial Audio in Extended Reality Applications		222		853–863	2021
Pelorson, X.; Van Hirtum, A.	SYNTHESIS OF THE HUMAN VOICE USING PHYSICAL MODELS AND REDUCED SET OF PARAMETERS					2024
Gentry, K. E.; Lewis, R. N.; Glanz, H.; Simões, P. I.; Nyári, Á. S.; Reichert, M. S.	Bioacoustics in cognitive research: Applications, considerations, and recommendations	Wiley Interdisciplinary Reviews: Cognitive Science	11	5		2020
Blauert, J.	Analysis and synthesis of auditory scenes				1–25	2005
Steinmetz, C. J.; Uhle, C.; Everardo, F.; Mitcheltree, C.; McElveen, J. K.; Jot, J.-M.; Wichern, G.	Audio Signal Processing in the Artificial Intelligence Era: Challenges and Directions	AES: Journal of the Audio Engineering Society	73	8	406–428	2025
Bank, B.; Sujbert, L.	Efficient modeling strategies for the geometric nonlinearities of musical instrument strings				691–696	2005

Hu, X.; Assaad, R. H.	A BIM-enabled digital twin framework for real-time indoor environment monitoring and visualization by integrating autonomous robotics, LiDAR-based 3D mobile mapping, IoT sensing, and indoor positioning technologies	Journal of Building Engineering	86			2024
Xi, Q.; Fu, Z.; Xue, M.-A.; Li, J.; Atroshchenko, E.; Zheng, J.	Numerical study of underwater acoustic radiation and propagation induced by structural vibration in ocean environments using FEM-BMSBM	Applied Ocean Research	153			2024
Chaplain, G. J.; Langfeldt, F.; Romero-García, V.; Jiménez, N.; Meng, Y.; Boulvert, J.; Groby, J. P.; Pagneux, V.; Moore, D. B.; Hibbins, A. P.; Sambles, J. R.; Starkey, T. A.; Popa, B.-I.; Zhang, Z.; Christensen, J.; Wen, X.; Li, J.; Fleury, R.; Wallen, S. P.; Haberman, M. R.; Hussein, M. I.; Memoli, G.; Fusaro, G.; D’Orazio, D.; Barbaresi, L.; Garai, M.; Chisari, L.; Mittal, P.; Qi, Z.; Subramanian, S.; Br?lé, S.; Enoch, S.; Guenneau, S.; Stoakes, R.; McLean, C.; Gardiner, A.	The 2024 acoustic metamaterials roadmap	Journal of Physics D: Applied Physics	58	43		2025
Wu, J.	Handbook of Contemporary Acoustics and its Applications					2016
Kumar, S. B.; Rangaiah, Y. P.; Singh, N.; Boob, N. S.; Srinivas, T.; AL-Zahra Jabbar, A. S. A.	Real-Time Revelations through Complex Machine Learning Techniques for Environmental Monitoring					2024
Bilbao, S.; Desvages, C.; Ducceschi, M.; Hamilton, B.; Harrison-Harsley, R.; Torin, A.; Webb, C.	Physical modeling, algorithms, and sound synthesis: The NESS project	Computer Music Journal	43	2-Mar	15–30	2019
Savioja, L.; Svensson, U. P.	Overview of geometrical room acoustic modeling techniques	Journal of the Acoustical Society of America	138	2	708–730	2015

Bilbao, S.; Perry, J.; Graham, P.; Gray, A.; Kavoussanakis, K.; Delap, G.; Mudd, T.; Sassoon, G.; Wishart, T.; Young, S.	Large-scale physical modeling synthesis, parallel computing, and musical experimentation: The ness project in practice	Computer Music Journal	43	2-Mar	31–47	2020
Döllinger, M.; Zhang, Z.; Schoder, S.; Šidlof, P.; Tur, B.; Kniesburgs, S.	Overview on state-of-the-art numerical modeling of the phonation process	Acta Acustica	7			2023
Tache, O.; Cadoz, C.	Generation of complex sound sequences using physical models with dynamical structures				1–8	2006
Renard-Julián, E. J.; Olmos, J. M.; García-Cascales, M. S.	A BIM-Oriented Framework for Integrating IoT-Based Air Quality Monitoring Systems Using the AllBIMclass Classification	Applied Sciences (Switzerland)	15	19		2025
Karjalainen, M.; Tolonen, T.; Välimäki, V.; Erkut, C.; Laurson, M.; Hiipakka, J.	An Overview of New Techniques and Effects in Model-based Sound Synthesis	International Journal of Phytoremediation	21	1	203–212	2001
Willemsen, S.; Bilbao, S.; Ducceschi, M.; Serafin, S.	Dynamic Grids for Finite-Difference Schemes in Musical Instrument Simulations				144–151	2021
Döbereiner, L.; Pirr?, D.	SPECULATIVE MACHINE LEARNING IN SOUND SYNTHESIS				515–522	2024
Rus, T.; Moldovan, R. P.; Mârza, C. M.; Corsiuc, G.; Iluțiu-Varvara, D.-A.	Data-driven environments: Evaluating IoT sensors and KNX protocol for monitoring indoor conditions in educational facilities	Frontiers in Built Environment	11			2025
Meza, G.	Exploring the potential of interactive Machine Learning for Sound Generation: A preliminary study with sound artists					2023
Coulby, G.; Clear, A. K.; Jones, O.; Godfrey, A.	Low-cost, multimodal environmental monitoring based on the Internet of Things	Building and Environment	203			2021
Schmidt, H.; Jensen, F. B.	Computational ocean acoustics: Advances in 3D ocean acoustic modeling		1495		3–15	2012
Picinali, L.	3D sound simulation over headphones				113–131	2009
Ducceschi, M.; Bilbao, S.	Non-iterative solvers for nonlinear problems: The case of collisions				17–24	2019

Leonard, J.; Villeneuve, J.	Mi-Gen?: An efficient and accessible mass-interaction sound synthesis toolbox				340–347	2019
Sakuma, T.; Sakamoto, S.; Otsuru, T.	Computational simulation in architectural and environmental acoustics: Methods and applications of wave-based computation		9,78443E+12			2014
Tarjano, C.; Pereira, V.	Neuro-Spectral Audio Synthesis: Exploiting Characteristics of the Discrete Fourier Transform in the Real-Time Simulation of Musical Instruments Using Parallel Neural Networks		11730 LNCS		362–375	2019
Soares, F.; Cochelin, B.; Freour, V.; Vergez, C.; Arimoto, K.	Nonlinear acoustics in brass instruments using 2D complex modal bases	Acta Acustica	9			2025
Sunder, K.; Jain, K.	Spatial Audio Technologies for Music, Gaming, and XR - A Primer					2022
Etter, Paul C.	Challenges in environmental acoustics		2		877–885	2000
Kliestik, T.; Kral, P.; Bugaj, M.; Durana, P.	Generative artificial intelligence of things systems, multisensory immersive extended reality technologies, and algorithmic big data simulation and modelling tools in digital twin industrial metaverse	Equilibrium. Quarterly Journal of Economics and Economic Policy	19	2	429–461	2024
Manocha, D.; Calamia, P.; Lin, M. C.; Savioja, L.; Tsingos, N.	Interactive sound rendering					2009
Chatziioannou, V.; Van Walstijn, M.	Energy conserving schemes for the simulation of musical instrument contact dynamics	Journal of Sound and Vibration	339		262–279	2015
Sallam, A.; Shahab, S.	Nonlinear Acoustic Holography With Adaptive Sampling	IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control	70	11	1516–1526	2023
Smith, J. O.	Virtual Acoustic Musical Instruments: Review and Update	Journal of New Music Research	33	3	283–304	2004
Feng, C.; Tian, X.	Analysis of Vocal Training Feedback Mechanisms Assisted by Artificial Intelligence				115–121	2026

Prinn, A. G.	A Review of Finite Element Methods for Room Acoustics	Acoustics	5	2	367–395	2023
Serra, X.	State of the art and future directions in musical sound synthesis				9–12	2007
Nicol, R.; Monfort, J.-Y.	Acoustic research for telecoms: bridging the heritage to the future	Acta Acustica	7	1		2023
Ji, J. C.; Luo, Q.; Ye, K.	Vibration control based metamaterials and origami structures: A state-of-the-art review	Mechanical Systems and Signal Processing	161			2021
Gupta, A.; Sharma, S.	Comparative Analysis of Sound Synthesis Methods and Development of a Compact Analog Synthesizer				198–207	2025
Widmer, G.; Rocchesso, D.; Välimäki, V.; Erkut, C.; Gouyon, F.; Pressnitzer, D.; Penttinen, H.; Polotti, P.; Volpe, G.	Sound and music computing: Research trends and some key issues	Journal of New Music Research	36	3	169–184	2007
Deng, M.; Menassa, C. C.; Kamat, V. R.	From BIM to digital twins: A systematic review of the evolution of intelligent building representations in the AEC-FM industry	Journal of Information Technology in Construction	26		58–83	2021
Kim, G.-H.; Kim, S.-B.; Moon, W.-K.; Hong, J.-Y.	Perception based sensory indicators in acoustic environment research: A scoping review	Journal of the Acoustical Society of Korea	45	1	41–54	2026
Wani, T. A.; Kumar, R. T.; Bansal, S.; Rashmi, P.; Thakuriya, K.; Reddy, B.; Deokar, A. A.	SMART SENSORS AND AI IN MUSICAL INSTRUMENT LEARNING	ShodhKosh: Journal of Visual and Performing Arts	6	3s	112–121	2025
Li, L.; Meng, W.; Bao, Y.	Bandgap metamaterials for vibration control: Theories, design, fabrication, and applications	Engineering Structures	347			2026
Masullo, M.; Maffei, L.	The Multidisciplinary Integration of Knowledge, Approaches and Tools: Toward the Sensory Human Experience Centres	Vibrations in Physical Systems	33	1		2022
Kiefer, C.	Towards Lightweight Architectures for Embedded Machine Learning in Musical Instruments				155–161	2022
Ronsse, L.; Cheenne, D. J.; Kadatz, S.	Using your ears: A novel way to teach acoustics		19			2013

Polfreman, R.	MULTI-MODAL INSTRUMENT: TOWARDS A PLATFORM FOR COMPARATIVE CONTROLLER EVALUATION				147–150	2011
Mohebbi-Kalkhoran, H.; Makris, N. C.; Ratilal, P.	Real-Time Detection, Bearing Estimation, and Whale Species Vocalization Classification From Passive Underwater Acoustic Array Data	IEEE Sensors Journal	24	22	37432–37444	2024
Taghizad, H.; Lemaire, M.; Massicotte, D.	Multi-rate real-time simulation: Techniques, models, frameworks, and challenges	Array	29			2026
Xia, R.; Guo, X.-W.; Li, C.; Liu, J.	Direct numerical simulation of acoustic wave propagation in ocean waveguides using a parallel finite volume solver	Ocean Engineering	281			2023
Zhao, Y.; Xia, X.; Togneri, R.	Applications of Deep Learning to Audio Generation	IEEE Circuits and Systems Magazine	19	4	19–38	2019
Chatterjee, S.; Klietik, T.; Rowland, Z.; Bugaj, M.	Immersive collaborative business process and extended reality-driven industrial metaverse technologies for economic value co-creation in 3D digital twin factories	Oeconomia Copernicana	16	1	125–161	2025
Liang, J.; Fan, S.; Chen, F.	XR Empowers a New City Landmark in Qingdao: Hi Metaverse		2116 CCIS		75–81	2024
Namgoong, M.; Kim, S.	Analysis of the audio engineering society’s research trend of the last four decades using the topic modeling of the AES publications					2023
Bech, S.; Zacharov, N.	Perceptual Audio Evaluation-Theory, Method and Application					2006
Pont, A.; Guasch, O.; Arnela, M.	Fast computational aeroacoustics using random distributions of Kirchhoff’s spinning vortices with application to sibilant sound generation					2019
Mason, J. D.	Organized mapping: Documenting a complex musical system					2007

Liu, S.; Niu, B.; Zhao, N.; Zong, G.; Zhao, X.; Li, L.	A novel event-triggered mechanism-based optimal safe control for nonlinear multi-player systems using adaptive dynamic programming	Journal of the Franklin Institute	362	11		2025
Schoder, S.; Weitz, M.; Maurerlehner, P.; Hauser, A.; Falk, S.; Kniesburges, S.; Döllinger, M.; Kaltenbacher, M.	Hybrid aeroacoustic approach for the efficient numerical simulation of human phonation	Journal of the Acoustical Society of America	147	2	1179–1194	2020
Gerner, M.	Her majesty, the queen of sounds: Cultural sustainability and heritage in organ craftsmanship and music	International Journal of Cultural Property	28	2	285–310	2021
Kuperman, W. A.; Song, H. C.	Integrating ocean acoustics and signal processing		1495		69–82	2012
Zhang, S.; Tian, J.; Banerjee, A.; Li, J.	An efficient porcine acoustic signal denoising technique based on EEMD-ICA-WTD	Mathematical Problems in Engineering	2019			2019
Kowol, P.; Nowak, P.; Di Nunzio, L.; Cardarilli, G. C.; Capizzi, G.; Sciuto, G. L.	Pipe Organ Design Including the Passive Haptic Feedback Technology and Measurement Analysis of Key Displacement, Pressure Force and Sound Organ Pipe	Applied System Innovation	7	3		2024
Allen, A.; Raghuvanshi, N.	Aerophones in Flatland: Interactive wave simulation of wind instruments		34	4		2015
Peters, P.	Crafting baroque sound: How the making of organ pipes matters artistically				125–136	2019
Ciaramella, A.; De Martino, S.; Tagliaferri, R.; De Lauro, E.; Falanga, M.	ICA for modelling and generating organ pipes self-sustained tones		1		261–265	2004
Kowol, P.; Nowak, P.; Lo Sciuto, G.	A Control Strategy for Mechatronic Action of a Pipe Organ Using a VCM Actuator	Electronics (Switzerland)	12	23		2023
Štafura, A.; Nagy, Š.	The upper lip as a voicing element of wooden organ pipes	Akustika	25	1	55–61	2016
Oertel, C. M.; Richards, A.	Music and materials: Art and science of organ pipe metal	MRS Bulletin	42	1	55–61	2017
Vaik, I.; Paál, G.	Flow simulations on an organ pipe foot model	Journal of the Acoustical Society of America	133	2	1102–1110	2013

Norman, L.; Chick, J. P.; Campbell, D. M.; Myers, A.; Gilbert, J.	Player control of 'brassiness' at intermediate dynamic levels in brass instruments	Acta Acustica united with Acustica	96	4	614–621	2010
Schlottke-Lakemper, M.; Yu, H.; Berger, S.; Meinke, M.; Schröder, W.	A fully coupled hybrid computational aeroacoustics method on hierarchical Cartesian meshes	Computers and Fluids	144		137–153	2017
Woolley, A.	Mechanical pipe organ actions and their interface with the player				411–415	2005
Takahashi, K.; Iwasaki, T.; Akamura, T.; Nagao, Y.; Nakano, K.; Kobayashi, T.; Takami, T.; Nishida, A.; Aoyagi, M.	Effective techniques and crucial problems of numerical study on flue instruments		19			2013
Rioux, Vincent	Sound quality of flue organ pipes. An interdisciplinary study on the art of voicing	Doktorsavhandlingar vid Chalmers Tekniska Hogskola		1680	1–76	2001
Jiang, J.; Liu, J.; Li, Z.; Zhang, T.; Yang, H.	Analysis on the mechanism of sound production and effects of musical flue pipe	Cognitive Computation and Systems	4	1	77–84	2022
Efkolidis, N.; Tsagaris, A.; Manavis, A.; Tzotzis, A.; Panagiotis, K.	Using 3d Digital Tools to Create a Digital Musical Instrument for A System Called HapticSOUND					2023
Außerlechner, H.; Angster, J.; Miklos, A.	Development of an adjustable pipe-foot model of a labial organ pipe				759–763	2008
Povilionis, G.	Observations on Metal Alloy and Types of Tongue and Shallot in Lithuanian Baroque Pipe Organs		23		187–220	2022
Meyer-Baese, U.; Odom, C.; Botella, G.; Meyer-Baese, A.	Independent component analysis algorithm FPGA design to perform real-time blind source separation		9496			2015
Bacciaglia, A.; Ceruti, A.; Liverani, A.	A design of experiment approach to 3D-printed mouthpieces sound analysis	Progress in Additive Manufacturing	6	3	571–587	2021
Lefebvre, A.; Scavone, G.	Wind instrument acoustic research in the computational acoustic modeling laboratory, McGill University		35	3	52–53	2007
Ridens, B. L.; Allison, T. C.; Simons, S.; Brun, K.	Modeling and mitigation of Acoustic Induced Vibration (AIV) in piping systems					2018

K?l?çarslan, D.; Kösterit, G.; Cigeroglu, E.	Frequency Domain Analysis of Aeroelastic Flutter Problem of Planar Structures Using Generalized Differential Quadrature Method				135–141	2025
	Advanced research instrumentation and facilities					2007
Štafura, A.; Nagy, Š.	Sound frequency spectrum of a wood organ pipe after the application of a second glue layer on the inner walls	Akustika	28	1	90–96	2017
d’Alessando , C.; Noisternig, M.; Le Beux, S.; Picinali, L.; Katz, B. F. G.; Jacquemin, C.; Ajaj, R.; Planes, B.; Strumel, N.; Delprat, N.	The ORA project: Audio-visual live electronics and the pipe organ				477–480	2009
Soares, L.; Aparo, E.; Almendra, R. A.	Design and the appropriation of 3D printing techniques in the management of an innovative product system in the field of wind musical instruments		22			2022
Ravina, E.; Silvestri, P.	Restoration of ancient pipes organs aided by experimental vibration and acoustic modal analyses		5		4016–4021	2010
Cavdir, D.	Embedding electronics in additive manufacturing for digital musical instrument design		2020-June		102–109	2020
Lombard, A.; Zheng, Y.; Buchner, H.; Kellermann, W.	TDOA estimation for multiple sound sources in noisy and reverberant environments using broadband independent component analysis	IEEE Transactions on Audio, Speech and Language Processing	19	6	1490–1503	2011
Han, Y.; Lee, K.	Detecting fingering of overblown flute sound using sparse feature learning	Eurasip Journal on Audio, Speech, and Music Processing	2016	1	1–10	2016
Simian, R.	3D-Printed Musical Instruments: Lessons Learned from Five Case Studies	Music and Science	6			2023
Qu, Q.; Zhang, H.; Luo, C.; Yu, R.	Robust control design for multi-player nonlinear systems with input disturbances via adaptive dynamic programming	Neurocomputing	334		1–10	2019

Ali, R.; Sadi, A. H. M. S.; Islam, A.; Rana, M. S.; Nasrin, S.; Shajol, S. A.	Machine Learning-Based Acoustic Signal Processing for Bowl Sound Analysis	International Journal of Artificial Intelligence and Machine Learning	4	2	9–22	2024
Bordoni, P.; Ody, P.; Kotus, J.; Kostek, B.	Sounding Mechanism of a Flue Organ Pipe—A Multi-Sensor Measurement Approach	Sensors	24	6		2024
Mihaela, M. A.; Gheorghe, G. I.; Luigi, G.; Daniel, C. C.; Octavian, D. G.	The control of the mechanical transmissibility for a M/K/C system with 1 DOF by replacing the dynamic absorber with an electro-mechanical system with adjustable electric parameters	Romanian Review Precision Mechanics, Optics and Mechatronics		44	105–111	2013
De Roeck, W.; Rubio, G.; Baelmans, M.; Sas, P.; Desmet, W.	The influence of flow domain modeling on the accuracy of direct and hybrid aeroacoustic noise calculations		1		251–271	2006
Stanzial, D.	On the intensimetric analysis and monitoring of flue organ pipes				641–646	2005
Howe, R.; Shahbazmohamadi, S.; Bass, R.; Singh, P.	Digital evaluation and replication of period wind instruments: The role of micro-computed tomography and additive manufacturing	Early Music	42	4	529–536	2014
Bruce Ralphin Rose, J.; Jinu, G. R.	A study on aeroelastic flutter suppression and its control measures -past and future	International Journal of Engineering and Technology	6	2	960–973	2014
Pati, R.; Pujari, A. K.; Gahan, P.; Kumar, V.	Independent component analysis: A review with emphasis on commonly used algorithms and contrast function	Computacion y Sistemas	25	1	97–115	2021
Kursell, J.; Peters, P.	Making Sound Present: Reenactment and Reconstruction in Historical Organ Building Practices				115–139	2025
Stanacevic, M.; Li, S.; Cauwenberghs, G.	Micropower mixed-signal VLSI independent component analysis for gradient flow acoustic source separation	IEEE Transactions on Circuits and Systems I: Regular Papers	63	7	972–981	2016
Außerlechner, H. J.; Trommer, T.; Angster, J.; Miklós, A.	Experimental jet velocity and edge tone investigations on a foot model of an organ pipe	Journal of the Acoustical Society of America	126	2	878–886	2009

Brendel, A.; Haubner, T.; Kellermann, W.	A Unifying View on Blind Source Separation of Convolutional Mixtures Based on Independent Component Analysis	IEEE Transactions on Signal Processing	71		816–830	2023
Węgrzyn, D. T.; Wrzeciono, P.; Wiczorkowska, A. A.	Predictive algorithms for determining blowing pressure in a pipe organ	Bulletin of the Polish Academy of Sciences: Technical Sciences	74	1		2026
Ziani, S.; Suchetha, M.; Ziani, L.	Enhancing Signal Quality: A Comparative Study of ICA and PCA in Denoising EMG and ECG Signals		1165		189–196	2024
Hacherai, Z. E.; Dequand, S.; Hallez, R.; Piccirillo, M.; Tournour, M.; Montavon, C.; Jones, I. P.; Tremblay, F.	Lighthill approach applied to industrial problems				225–230	2002
Boutin, H.; Smith, J.; Wolfe, J.	'Warming up' a wind instrument: The time-dependent effects of exhaled air on the resonances of a trombone	Journal of the Acoustical Society of America	148	4	1817–1823	2020
Raihan, G. A.; Chakravarty, U. K.	Fluid-Structure Interaction Model of a Wind Turbine Blade		4			2023
Ikoga, S.; Onomata, T.; Tabata, R.; Iwagami, S.; Kobayashi, T.; Takahashi, K.	NUMERICAL STUDY ON A THREE-DIMENSIONAL MODEL OF A FLUE ORGAN PIPE: RELATIVE PHASE BETWEEN PIPE AND FOOT, AND STABILITY					2023
Li, S.; Karney, B. W.; Liu, G.	FSI research in pipeline systems - A review of the literature	Journal of Fluids and Structures	57		277–297	2015
Boutin, H.; Le Conte, S.; Vaiedelich, S.; Fabre, B.; Le Carrou, J.-L.	Acoustic dissipation in wooden pipes of different species used in wind instrument making: An experimental study	Journal of the Acoustical Society of America	141	4	2840–2848	2017
Nederveen, C. J.; Dalmont, J.-P.	Pitch and level changes in organ pipes due to wall resonances	Journal of Sound and Vibration	271	1-Feb	227–239	2004
Brewster, J. D.	Lattice-Boltzmann simulations of three-dimensional fluid flow on a desktop computer	Analytical Chemistry	79	7	2965–2971	2007
Tournour, M.; Atalla, N.	Pseudostatic corrections for the forced vibroacoustic response of a structure-cavity system	Journal of the Acoustical Society of America	107	5 I	2379–2386	2000

Göken, J.	Temperature-dependent damping of the tonewood spruce	Journal of Alloys and Compounds	856			2021
Pitsch, S.; Holmberg, S.; Angster, J.	Ventilation system design for a church pipe organ using numerical simulation and on-site measurement	Building and Environment	45	12	2629–2643	2010
Li, M.; Yan, H.; Wang, L.	Data-driven model reduction for pipes conveying fluid via spectral submanifolds	International Journal of Mechanical Sciences	277			2024
Tsuchida, J.; Fujisawa, T.; Yagawa, G.	Direct numerical simulation of aerodynamic sounds by a compressible CFD scheme with node-by-node finite elements	Computer Methods in Applied Mechanics and Engineering	195	13-16	1896–1910	2006
Xu, J.; Chung, K. W.; Chan, H. S. Y.	Co-dimension 2 bifurcations and chaos in cantilevered pipe conveying time varying fluid with three-to-one internal resonances	Acta Mechanica Sinica	16	3	245–255	2003
Kraxberger, F.; Näger, C.; Laudato, M.; Sundström, E.; Becker, S.; Mihaescu, M.; Kniesburges, S.; Schoder, S.	On the Alignment of Acoustic and Coupled Mechanic-Acoustic Eigenmodes in Phonation by Supraglottal Duct Variations	Bioengineering	10	12		2023
Picó, R.; Gilbert, J.; Gautier, F.	Study of the wall vibration effect on the oscillations of a distorted wind instrument		4		3093–3098	2005
He, X.; Doolen, G. D.; Clark, T.	Comparison of the lattice Boltzmann method and the artificial compressibility method for Navier-Stokes equations	Journal of Computational Physics	179	2	439–451	2002
Yoshikawa, S.; Tashiro, H.; Sakamoto, Y.	Experimental examination of vortex-sound generation in an organ pipe: A proposal of jet vortex-layer formation model	Journal of Sound and Vibration	331	11	2558–2577	2012
Yoshikawa, S.	A pictorial analysis of jet and vortex behaviours during attack transients in organ pipe models	Acustica	86	4	623–633	2000
Bocanegra, J.; Borelli, D.	Review of acoustic hysteresis in flute-like instruments					2019
Li, Q.; Yu, Y.; Tang, S.	Multiphase lattice Boltzmann method and its applications in phase-change heat transfer	Kexue Tongbao/Chinese Science Bulletin	65	17	1677–1693	2020

Yang, Z.; Zhao, X.; Zhang, S.; Chen, C.-P.	Fluid-structure interaction for flutter predictions in transonic and supersonic flows		1	PARTS A	C	D
Li, W.; Wang, W.-Q.; Yan, Y.; Yu, Z.-F.	A strong-coupled method combined finite element method and lattice Boltzmann method via an implicit immersed boundary scheme for fluid structure interaction	Ocean Engineering	214			2020
Axner, L.; Hoekstra, A. G.; Jeays, A.; Lawford, P.; Hose, R.; Sloot, P. M.	Simulations of time harmonic blood flow in the Mesenteric artery: comparing finite element and lattice Boltzmann methods.	Biomedical engineering online	8		23	2009
Angster, J.; Wik, T.; Taesch, C.; Miklós, A.; Sakamoto, Y.	Experiments on the influence of pipe scaling parameters on the sound of flue organ pipes				603–609	2005
Pałczyński, T.	Influence of Air Temperature on Dynamic Properties of Pipes Supplied with Pulsating Flow		249		291–302	2018
Blades, E. L.; Cornish, A.	Aeroelastic stability predictions of a business jet landing gear door using high-fidelity fluid-structure interaction tools					2015
Hao, M.-Y.; Ding, H.; Mao, X.-Y.; Chen, L.-Q.	Multi-harmonic resonance of pipes conveying fluid with pulsating flow	Journal of Sound and Vibration	569			2024
Taylor, A. J.	Pipe organ reservoir resonance.		41		319–326	2019
Li, Q.; Liu, W.; Lu, K.; Yue, Z.	Nonlinear Parametric Vibration of a Fluid-Conveying Pipe Flexibly Restrained at the Ends	Acta Mechanica Solida Sinica	33	3	327–346	2020
Panda, L. N.; Kar, R. C.	Nonlinear dynamics of a pipe conveying pulsating fluid with combination, principal parametric and internal resonances	Journal of Sound and Vibration	309	3-May	375–406	2008
De Lauro, E.; De Martine, S.; Esposito, E.; Sarno, G.; Tomasini, E. P.	Modellization of mechanical vibrations in organ pipes by independent component analysis		6345			2006
Ikoga, S.; Onomata, T.; Tabata, R.; Iwagami, S.; Kobayashi, T.; Takahashi, K.	Numerical study on role of foot of a flue organ pipe: relative phases in oscillations among pipe, foot and jet	Acta Acustica	8			2024

Rucz, P.; Angster, J.; Augusztinovicz, F.; Máté Márton, V.; Miklós, A.	Air jet and edge tone simulation in an organ pipe foot model		7		5971–5980	2013
Wu, Z.; Guo, L.	Accuracy improvement of immersed boundary-lattice Boltzmann and finite element method by iterative velocity correction	Physics of Fluids	34	10		2022
Li, Q.; Liu, W.; Zhang, Z.; Yue, Z.	Parametric resonance of pipes with soft and hard segments conveying pulsating fluids	International Journal of Structural Stability and Dynamics	18	10		2018
Gautier, F.; Nief, G.; Gilbert, J.; Dalmont, J. P.	Vibro-acoustics of organ pipes-Revisiting the Miller experiment (L)	Journal of the Acoustical Society of America	131	1	737–738	2012
Wang, Z.; Wu, H.; Shi, Y.-F.; Yang, J.-G.; Cai, J.	Fluid-structure interaction and flutter analysis of compressor blade based on CFD/CSD	Hangkong Dongli Xuebao/Journal of Aerospace Power	26	5	1077–1084	2011
Luo, L.; Bochenina, K.; Abuhay, T. M.; Dorzhu, N.; Kamps, G.; Kovalchuk, S.; Krzhizhanovskaya, V.; Paszynski, M.; de Mulatier, C.; Dongarra, J.; Sloom, P. M. A.	Evolution of the computational science community: The dynamics of topics and collaborations in 24 years of ICCS and JoCS publications	Journal of Computational Science	89			2025
Guillemain, P.; Kergomard, J.; Voinier, T.	Real-time synthesis of clarinet-like instruments using digital impedance models	Journal of the Acoustical Society of America	118	1	483–494	2005
Bakunić, E.; Zhang, Y.; Fareed, B.; Khzouz, M.; Ali, S.; Mostafa, N.; Yaqoob, H.; Kalinowska-Wichrowska, K.	Artificial intelligence in fluid dynamics and multiphase flow systems				257–284	2025
Hinrichsen, A.; Bovermann, T.	Post-DMI musical instruments		04-06-October-2016		124–131	2016
Karn, A.; Anand, R.; Kataria, A.; Donga, R. K.; Agarwal, N.; Singh, V. P.	Playing flute without lips? Tones of music lost in time:An investigation of the indigenous Bastar flutes of India	International Journal of Aeroacoustics	22	1-Feb	110–130	2023
Gaillard, A.; Koehl, V.; Gazengel, B.	Theoretical and experimental studies about single cane reeds: a review	Acta Acustica	8			2024
Murad, J.	Harnessing the power of the cloud-computational fluid dynamics with simscale		1			2021

Fan, S.; Ge, Y.; Li, B.; Liu, P.; Liu, X.	Advancements in Microfluidic Organ-on-a-chip for Oral Medicine	International Dental Journal	75	5		2025
Darabundit, C. C.; Scavone, G.	Discrete port-Hamiltonian system model of a single-reed woodwind instrument	Frontiers in Signal Processing	5			2025
Rucz, P.; Augusztinovicz, F.; Fiala, P.	Simulation of organ pipes' acoustic behavior by means of various numerical techniques		3		1742–1749	2009
Marini, D.; Polychronopoulos, S.; Bakogiannis, K.; Kouroupetoglou, G.	A unified comparative approach for wind instrument physical modeling: Finite element vs digital signal processing methods					2021
Zhao, D.	Thermoacoustic Combustion Instability Control: Engineering Applications and Computer Codes					2023
		Thermoacoustic {Combustion} {Instability} {Control}: {Engineering} {Applications} and {Computer} {Codes}		Elsevier		
Hergert, F.; Hale, P.	A Review of Technical Inventions to include deep Bass Tones into Pipe Organs despite Space Constraints		49	1		2022
Justo-Estebarez, A.; Herrera, L. K.; Duran, A.; Sigüenza, B.; Jimenez De Haro, M. C.; Laguna, O.; Justo, A.	Analysis of the restoration of an historical organ: The case study of the Cavaillé-Coll organ of la Merced Church in Burgos, Spain	Studies in Conservation	57	1	21–28	2012
Bilbao, S.	Direct simulation for wind instrument synthesis				145–152	2008
Kob, M.	Influence of wall vibrations on the transient sound of a flue organ pipe	Acustica	86	4	642–648	2000
Yu, H.; Xu, Y.; Liu, H.; Wang, J.	Study on Numerical Simulation of Cavitation Jet Flow for Two-chamber Self-vibrating Pulse Nozzles	Jixie Kexue Yu Jishu/Mechanical Science and Technology for Aerospace Engineering	42	4	585–591	2023
Miklós, A.; Angster, J.; Pitsch, S.; Rossing, T. D.	Reed vibration in lingual organ pipes without the resonators	Journal of the Acoustical Society of America	113	2	1081–1091	2003

Abel, M.; Bergweiler, S.; Gerhard-Multhaupt, R.	Synchronization of organ pipes: Experimental observations and modeling	Journal of the Acoustical Society of America	119	4	2467–2475	2006
Hruška, V.; Dlask, P.	Connections between organ pipe noise and shannon entropy of the airflow: Preliminary results	Acta Acustica united with Acustica	103	6	1100–1105	2017
Ukshini, E.; Dirckx, J. J. J.	Three-dimensional vibration patterns of alto saxophone reeds measured on different mouth-pieces under mimicked realistic playing conditions	Journal of the Acoustical Society of America	150	5	3730–3746	2021
Hwu, C.	Boundary Element Analysis		267		339–448	2021
Krzyzaniak, M.	How to build pipe organ robots				121–128	2022
Xu, J.; Li, B.; Zhao, Y.; Xue, S.	Sound source localization based on data and neural network model		2816	1		2024
Rogal, J.; Schlünder, K.; Loskill, P.	Developer’s Guide to an Organ-on-Chip Model	ACS Biomaterials Science and Engineering	8	11	4643–4647	2022
Tian, W.; Yao, L.; Li, L.	A Coupled Smoothed Finite Element-Boundary Element Method for Structural-Acoustic Analysis of Shell	Archives of Acoustics	42	1	49–59	2017
Kumar, D.; Nadda, R.; Repaka, R.	Advances and challenges in organ-on-chip technology: toward mimicking human physiology and disease in vitro	Medical and Biological Engineering and Computing	62	7	1925–1957	2024
Adachi, S.; Angster, J.; Miklos, A.	Mode transition of a flue organ pipe				747–752	2008
Vasconez Martinez, M. G.; Frauenlob, M.; Rothbauer, M.	An update on microfluidic multi-organ-on-a-chip systems for reproducing drug pharmacokinetics: the current state-of-the-art	Expert Opinion on Drug Metabolism and Toxicology	20	6	459–471	2024
Stavrogiannis, C.; Sofos, F.; Karakasidis, T. E.	Machine Learning-Enhanced Molecular Dynamics: Current State, Challenges and Perspectives	Archives of Computational Methods in Engineering				2026
Domenighini, P.; Buratti, C.	CFD analysis of thermal fluid dynamic parameters inside an organ pipe: influence of air temperature variations		2685	1		2024
Manoj, K.; Pawar, S. A.; Kurths, J.; Sujith, R. I.	Rijke tube: A nonlinear oscillator	Chaos	32	7		2022

Manchanda, R.; Khalid, M.; Arora, P.	A Bibliometric Analysis of Computational Intelligence: Trends, Challenges, and Future Directions in AI and Machine Learning Research		1601 LNNS		185–199	2026
Spolska, O.; Kyreia, M.; Hryhoruk, N.; Saldan, S.; Riabov, I.	Pneumatics, acoustics and digital sound: The organ in the history of science and technology	History of Science and Technology	15	2	472–498	2025
Zhang, Y.; Wang, L.; Zhao, Q.; Gong, J.; Tian, Y.; Liu, Y.; Chen, S.; Zhao, X.; Yang, Q.; Zhao, Y.	Mapping the trajectory of organoid research: Insights into complex models and advanced technologies	Engineered Regeneration	6		264–276	2025
Baretzky, B.; Friesel, M.; Petelin, A.; Shotanov, A.; Straumal, B.	Reconstruction of brass for tongues and shallots from Baroque organs		258-260		397–402	2006
Su, W.; Zhao, Y.; Yeboah, S.; Li, X.; Wang, L.	Numerical models for organ-on-a-chip: A systematic review and analyses	Biomicrofluidics	19	4		2025
Runchal, A. K.	Evolution of CFD as an engineering science. A personal perspective with emphasis on the finite volume method	Comptes Rendus - Mecanique	350			2022
Cho, K.	Computational science and the search for dark matter	New Physics: Sae Mulli	66	8	950–956	2016
Wu, X.-Y.; Zhang, Y.-Q.; Xu, Z.-Q.; Zhao, S.; Li, G.-S.; Tian, S.-C.; Tan, Y.-W.; Peng, K.-W.	Structure optimization of the organ-pipe cavitating nozzle and its erosion ability test on hydrate-bearing sediments	Petroleum Science	20	2	1104–1118	2023
Puranik, N.; Scavone, G.	CLAMPED BAR MODEL FOR FREE REEDS					2023
Gilbert, J.; Maugeais, S.; Vergez, C.	Minimal blowing pressure allowing periodic oscillations in a simplified reed musical instrument model: Bouasse-Benade prescription assessed through numerical continuation	Acta Acustica	4	6		2020
Miklós, A.; Angster, J.	Properties of the sound of flue organ pipes	Acustica	86	4	611–622	2000

Pawlik, L.; Wilk-Jakubowski, J. L.; Frej, D.; Wilk-Jakubowski, G.	Applications of Computational Mechanics Methods Combined with Machine Learning and Neural Networks: A Systematic Review (2015–2025)	Applied Sciences (Switzerland)	15	19		2025
Saravanabalaji, M.; Sivakumaran, N.; Sankarnaraynan, S.	Dynamic Compensation of Acoustic Resonance for Water Flow System	Journal of The Institution of Engineers (India): Series B	100	5	405–415	2019
Welander, P.	Ethernet connectivity for pipe organs	Control Engineering	55	4	M1–M4	2008
Esling, P.; Masuda, N.; Bardet, A.; Despres, R.; Chemla-Romeu-Santos, A.	Flow synthesizer: Universal audio synthesizer control with normalizing flows	Applied Sciences (Switzerland)	10	1		2020
Hruška, V.; Dlask, P.	Investigation of the sound source regions in open and closed organ pipes	Archives of Acoustics	44	3	467–474	2019
Durkin, R.; Martin, D.	ON ORGANOLOGY: Taxonomy and Transdisciplinarity				285–294	2023
Fang, Z.; Hou, W.; Fan, S.; Guo, X.; Chen, Y.	Coherent structure analysis of cavitation waterjets using dynamic mode decomposition	Physics of Fluids	36	3		2024
Xu, J.; Wang, F.	Cardiac Mechano-Electrical-Fluid Interaction: A Brief Review of Recent Advances	Eng	6	8		2025
Ishida, F.; Samejima, T.	Physical modeling of wind instruments involving three-dimensional radiated sound fields	Acoustical Science and Technology	44	3	247–258	2023
Coman, E.	IVR systems used in call center management: a scientometric analysis of the literature	Frontiers in Computer Science	7			2025
D'Alessandro, C.; Noisternig, M.	Of Pipes and Patches: Listening to augmented pipe organs	Organised Sound	24	1	41–53	2019
Shaaban, M.; Mohany, A.	Passive control of flow-excited acoustic resonance in rectangular cavities using upstream mounted blocks	Experiments in Fluids	56	4		2015

Schlick, T.; Portillo-Ledesma, S.; Blaszczyk, M.; Dalessandro, L.; Ghosh, S.; Hackl, K.; Harnish, C.; Kotha, S.; Livescu, D.; Masud, A.; Matouš, K.; Moyeda, A.; Oskay, C.; Fish, J.	A MULTISCALE VISION—ILLUSTRATIVE APPLICATIONS FROM BIOLOGY TO ENGINEERING	International Journal for Multiscale Computational Engineering	19	2	39–73	2021
Adachi, S.	Principles of sound production in wind instruments	Acoustical Science and Technology	25	6	400–405	2004
De Chiara, F.; Fernández-Costa, J. M.; Ramón-Azcón, J.; Grochala, J.; Paleczek, A.; Grochala, D.; Rydosz, A.	Major on-chip applications				187–265	2023
Csonka, D.; Storc, T.; Kaszás, A.; Forberger, Á.; Várady, G.	Surrogate Modelling and Simulation Approaches for Renal Artery Haemodynamics: Balancing Symmetry in Computational Cost and Accuracy	Symmetry	17	10		2025
Taylor, A. J.	The effects of centrifugal blowers and reservoir resonance on organ pipe flutter		40		315–322	2018
Warszewa, A.; Böhm, M.; Sawodny, O.; Tarín, C.	A port-Hamiltonian approach to modeling the structural dynamics of complex systems	Applied Mathematical Modelling	89		1528–1546	2021
Woolley, A.	Transient variation in mechanical action and electric action pipe organs		19			2013
Hu, J.; Kang, Y.; Shi, H.; Wang, X.; Hu, Y.; Li, D.	Experimental and numerical investigation on the cavitation erosion effect and resonance effect of submerged water jets	Ships and Offshore Structures	21	3	355–367	2026
Al-Ogbi, S. A.; Felemban, M.	Solving Computational Fluid Dynamic Problems Using Quantum Computing					2025
Baretzky, B.; Friesel, M.; Straumal, B.	Reconstruction of historical alloys for pipe organs brings true baroque music back to life	MRS Bulletin	32	3	249–255	2007
Zhang, D.; Anjum, T.; Chu, Z.; Cross, J. S.; Ji, G.	Simulation of multiphase flow with thermochemical reactions: A review of computational fluid dynamics (CFD) theory to AI integration	Renewable and Sustainable Energy Reviews	221			2025

Carlsson, T.	On dynamic behaviour of wind systems for pipe organs	Doktorsavhandlingar vid Chalmers Tekniska Hogskola		1809		2002
Fang, Z.; Hou, W.; Xu, Z.; Guo, X.; Zhang, Z.; Shi, R.; Yao, Y.; Chen, Y.	Large Eddy Simulation of Cavitation Jets from an Organ-Pipe Nozzle: The Influence of Cavitation on the Vortex Coherent Structure	Processes	11	8		2023
Howard, D. M.	Developmental History of the Vocal Tract Organ	Journal of Voice	39	1	57-64	2025